

Customer Requirement Specification

ACME Mobile LPDDR5X Memory Subsystem

Document No. CRS-ACME-0042, Rev. B — Synthetic sample

1. Overview

This specification lists the requirements of a fictional customer for an LPDDR5X x16 memory subsystem used in a handheld product. Requirement IDs of the form REQ-XXX-nnn are mandatory unless stated otherwise.

1.1 Purpose

The memory vendor shall confirm each requirement in the compliance matrix (Table 1) within 10 business days.

2. Requirements

2.1 Power Requirements

Table 1 — Compliance Matrix

Req. ID	Requirement	Target	Priority
REQ-PWR-001	Self refresh current at 25 °C	≤ 0.8 mA	High
REQ-PWR-002	Deep sleep power	≤ 5 mW	High
REQ-PWR-003	VDD2H supply range	1.01 V to 1.12 V	Medium
REQ-TMP-001	Operating temperature	-40 °C to 95 °C	High
REQ-SIG-001	Max data rate	8533 Mbps	Medium

Requirements marked High shall be verified by test reports. REQ-SIG-001 applies to every DQ[15:0] lane.

2.2 System Integration

Figure 1 shows the integration of the memory in the application platform. The PMIC supplies VDD2H = 1.05 V and the thermal sensor reports over I2C at 400 kHz.

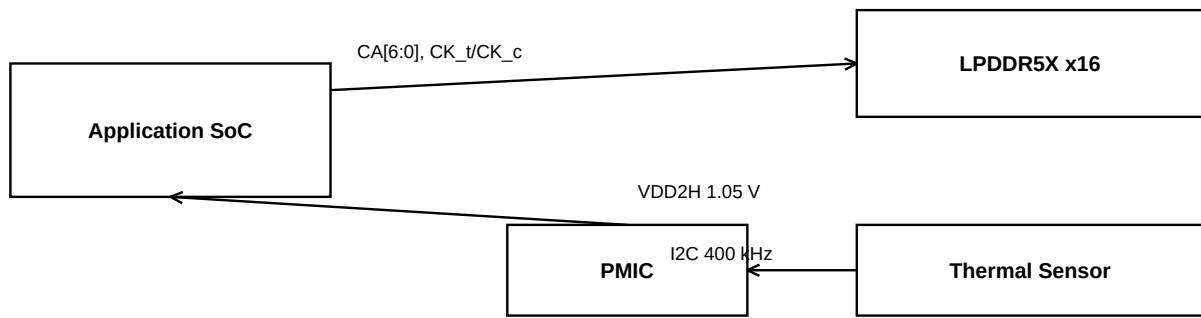


Figure 1 — System Block Diagram

2.3 Qualification Schedule

The qualification milestones are listed in Table 2.

Table 2 — Qualification Milestones

Milestone	Date	Owner
Engineering samples	2026-11-15	Vendor
Reliability report	2027-01-31	Vendor
Customer qualification	2027-03-15	ACME

3. Glossary

CRS: Customer Requirement Specification, this document.

PMIC: Power management integrated circuit supplying VDD1, VDD2H and VDDQ.

ES: Engineering sample, built on a pre-production process.

Rev.: Document revision; Rev. B supersedes Rev. A.

Mbps: Megabits per second per pin.

TBD: To be defined in the next revision.